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A three decade assessment of climate-associated changes in forest composition across the north-eastern USA

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Summary

1. Climate-associated changes in forest composition have been widely reported, particularly where changes in abiotic conditions have resulted in high mortality of sensitive species and have disproportionately favoured certain species better adapted to these newer conditions. In the north-eastern USA and south-eastern Canada, few studies have examined climate-related influences associated with forest composition, and none have considered broad-scale changes over a long temporal (>25 years) period.

2. We used US Forest Service Forest Inventory and Analysis data from 1983 to 2014 across four north-eastern states (Maine, New Hampshire, New York and Vermont) to assess temporal and spatial changes in the occurrence and abundance of American beech *Fagus grandifolia* Ehrh, sugar maple *Acer saccharum* L., red maple *Acer rubrum* L. and birch *Betula* spp. saplings. We also tested the effects of biotic and abiotic factors on the distribution of the four studied deciduous species over the entire period examined.

3. Occurrence and abundance of American beech have increased substantially over the past three decades, whereas the occurrence and abundance of three other deciduous species have decreased in all ecological provinces of the north-eastern USA, except the Midwest Broadleaf ecological province. Consequently, a clear shift in species composition is currently underway in the beech-maple-birch (BMB) forests of the north-eastern USA, with uncertain consequences for future ecosystem structure and function.

4. In the studied region and over the entire period examined, the distribution of increased occurrence and abundance of beech relative to the three other deciduous species were associated with higher temperature and precipitation as well as higher conspecific basal area and dead tree basal area.

5. *Synthesis and applications*. The change from beech-maple-birch forests to more beech-dominated forests with beech encroachment to new forest areas across the north-eastern USA may continue if higher intensity harvesting and disturbances (i.e. large-scale canopy openings) do not occur. This would be a significant management concern as beech is associated with a widespread bark disease, is commercially less desirable, and can limit natural regeneration from other species. Our results emphasize the need for management strategies such as higher intensity harvesting methods, vegetation control and limiting browsing pressure to reduce beech dominance.

Key-words: abundance, American beech *Fagus grandifolia*, climate change, distribution, forest regeneration, north-eastern USA, northern hardwoods, temporal and spatial shifts in forest composition

Introduction

In recent decades, climate-associated shifts in forest composition have been increasingly reported (e.g. Rigling *et al.* 2013; Bernhardt-Römermann *et al.* 2015; Barros *et al.* 2017). The key factors identified have included elevated temperature, increased atmospheric CO_2 concentration and greater frequency of prolonged drought (Hansen *et al.* 2001; Cavin & Jump 2017). In forest ecosystems, elevated temperature has caused upward (altitudinal) and

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poleward (latitudinal) shifts of the tree line (Devi *et al.* 2008; Harsch *et al.* 2009), as well as changed the forest composition below the tree line (Lenoir *et al.* 2010; Bertrand *et al.* 2011). Shifts in species composition can cause plausible changes in ecosystem functioning, including biodiversity and productivity (Bellard *et al.* 2012; Bernhardt-Römermann *et al.* 2015). Assessing the effects of such changes on forest ecosystems is crucial, as they provide important services including fibre production, carbon sequestration, and habitat for key wildlife and plant species (Boisvenue & Running 2006; Manning, Gibbons & Lindenmayer 2009).

An important question in the debate on climate change and species-level adaptation is whether or not species will be able to adapt fast enough to maintain the rapid pace of change in environmental conditions (Bellard *et al.* 2012). Species composition of a forest ecosystem is generally driven by species-specific physiological adaptation to local abiotic and biotic environments (Grubb 1977). Species-level adaptation to climate change has been widely reported (e.g. Parmesan 2006; Lenoir & Svenning 2015). For example, a species can track appropriate conditions spatially through dispersal (Parmesan 2006), temporally by modifying seasonality behaviours such as phenotypic activities (Richardson *et al.* 2006), or cope with changing conditions by modifying species-specific physiological thresholds (Perry, Oren & Hart 2008). Drastic changes in biotic and/or abiotic environments can often result in high mortality of sensitive species, which can disproportionately favour species that are better adapted to the altered conditions (Allen *et al.* 2010). Therefore, climate change may favour the encroachment of dominant species (better adapted to changed conditions) in the forest landscape through various stochastic neutral processes such as extinction, immigration and speciation (Hubbell 2001), as well as driving shifts in coexistence mechanisms among species, and ultimately changing the community composition (Fournier *et al.* 2016).

The north-eastern region of USA has warmed by 0·25°C per decade since 1970 (Hayhoe *et al.* 2007). This warming is associated with an extended growing season, increased winter precipitation, and greater deviation in annual temperature (Hayhoe *et al.* 2007). In the north-east, beech-maple-birch (BMB) forests (≥80% of overstorey basal area is dominated by American beech *Fagus grandifolia* Ehrh., sugar maple *Acer saccharum* Marsh., red maple *Acer rubrum* L. and birch species *Betula* spp.) currently (2010–2014) covers 26% of the total forestland area (https://www.fia.fs.fed.us/). In recent years, several studies have reported increasing dominance of American beech relative to sugar maple, red maple and birch species in the understories of the northern hardwood forests (e.g. Hane 2003; Duchesne & Ouimet 2009). American beech has shown to be more efficient relative to sugar maple in assimilating carbon under elevated atmospheric CO_2 (Reid & Strain 1994). Moreover, Fang & Lechowicz (2006) demonstrated that elevated temperature, particularly during the growing season, has been favouring the expansion of the natural range of American beech. In addition, increased soil acidification has also benefited the understorey dominance of beech over sugar maple (Duchesne & Ouimet 2009), and the seedling abundance of sugar maple has been found to be negatively associated with the increased N deposition (Patterson *et al.* 2012).

In addition to climatic and other abiotic factors, the dominance of beech in forests of the north-eastern USA has also been attributed to important biotic factors including (i) extreme shade tolerance (Canham 1989), (ii) ability to regenerate vegetatively (Beaudet & Messier 2008), (iii) lower palatability to browsers (Tripler *et al.* 2005), and (iv) natural mortality due to insects and diseases (e.g. beech-bark disease) (Wagner *et al.* 2010). In these forests, diseased beech has low commercial value than other deciduous species, including sugar maple, red maple and birch species (Nyland *et al.* 2006). Therefore, the increasing beech dominance may significantly reduce the future diversity, productivity and value of northern hardwood forest stands.

The objectives of this study were to evaluate the encroachment (frequency of occurrence or colonization of new areas), absolute and relative abundance of saplings of four major deciduous tree species in forests of the north-eastern USA during the past three decades, and assess the effects of key biotic and abiotic factors on the distribution of sapling occurrence and abundance across the study region. In this context, we tested three hypotheses: (1) the encroachment (i.e. frequency of occurrence) of beech saplings into new areas has increased over the past three decades in association with the reduction in other associated deciduous species (sugar maple, red maple and birches) with a greater shift towards beech dominance in mountainous forest areas than in forests at lower elevations (Wason & Dovciak 2017), (2) in BMB forests (≥80% of overstorey basal area occupied by hardwoods) of north-eastern USA, the absolute and relative abundance of beech saplings has increased over the past three decades in association with the reduction in other associated deciduous species, and (3) abiotic factors (e.g. past temperature and precipitation) will be superior to biotic factors (e.g. overstorey basal area) in explaining the distribution of sapling occurrence, and absolute and relative abundance of the four deciduous species (Thuiller, Araujo & Lavorel 2004), with elevated temperature, greater precipitation and higher deviation from average conditions (variability in temperature and precipitation) expected to be associated with higher occurrence and abundance of beech relative to the other deciduous species (Fang & Lechowicz 2006; Wason & Dovciak 2017).

Materials and methods

STUDY AREA

The study area covers all forests of Maine, New Hampshire, Vermont and New York, which represents four distinct ecological provinces of this region (McNab *et al.* 2007): (i) Adirondack-New England Mixed Forest-Coniferous Forest-Alpine Meadow, (ii) North-eastern Mixed Forest, (iii) Eastern Broadleaf Forest and (iv) Midwest Broadleaf Forest. For simplicity, we have termed these four ecological provinces as (i) Alpine Mixed, (ii) North-eastern Mixed, (iii) Eastern Broadleaf and (iv) Midwest Broadleaf. The Forest Inventory and Analysis (FIA) database of USDA Forest Service (http://www.fia.fs.fed.us) was used for this analysis. A description of the study area, ecological provinces and the FIA data are given in Appendices S1–S5, Supporting Information.

ANALYTICAL APPROACH

For this study, all FIA plots from 1983 to 2014 available for Maine, New Hampshire, New York and Vermont were used. FIA plots were remeasured on a periodic basis with a general shift from 10- to 5-year measurement intervals in 2000. A total of 68 411 plots were used for the analysis.

Response variables

We characterized species-level encroachment into new areas (frequency of occurrence) by the presence or absence of a sapling (Thuiller *et al.* 2014). We modelled (i) presence or absence, (ii) absolute abundance (stems per ha) and (iii) relative abundance (relative to total) of saplings (2·5–12·69 cm at DBH) of four major deciduous species of north-eastern USA including American beech, sugar maple, red maple and birch species. For birch, we combined all available local species (*Betula alleghaniensis*, *Betula lenta*, *Betula nigra*, *Betula papyrifera* and *Betula populifolia*). We considered the entire dataset (n = 68 411) to analyse the sapling occurrence (presence or absence), whereas only BMB plots were used to analyse the absolute and relative sapling abundance of those four species. Number of sample plots considered for each objective (i.e. frequency of occurrence, absolute and relative abundance) is provided in the Appendix S6.

Explanatory variables

We used time (calendar year of measurement), ecological province (four levels) and species (four levels) to address the first and second hypotheses, that sapling frequency of occurrence (presence or absence), and absolute and relative abundance vary across space and time. For the third hypothesis, we used a range of biotic and abiotic factors (Table 1) as well as species to evaluate whether the spatial distribution of sapling occurrence and abundance were associated with important biotic and abiotic factors. We used the mean annual temperature, total mean annual precipitation, mean annual vapour pressure, mean annual growing degree days and mean annual frost free days as abiotic factors. In addition, we quantified the annual variability [deviation from long-term (1980–2014) to measurement year] of temperature, precipitation and vapour pressure (Table 1). We considered

Table 1. Summary statistics (mean ± 1SD) of biotic and abiotic explanatory variables considered in the analysis

Predictors	Ecological provinces: Alpine mixed	North-eastern mixed	Eastern broadleaf	Midwest broadleaf
Biotic				
Live overstorey basal area (m^2 ha^{-1})	27·2 ± 30·8	25·5 ± 30·8	28·0 ± 31·7	18·3 ± 21·3
Live overstorey density (stems per ha)	578 ± 654	596 ± 797	505 ± 568	298 ± 337
% of live beech basal area (% of total)	7·7 ± 19·4	4·6 ± 15·7	2·9 ± 12·0	2·4 ± 12·1
% of live sugar maple basal area (% of total)	14·6 ± 28·0	8·5 ± 22·2	5·7 ± 17·8	11·1 ± 26·3
% of live red maple basal area (% of total)	11·8 ± 22·1	14·4 ± 25·5	17·1 ± 27·6	13·1 ± 28·5
% of live birch basal area (% of total)	15·1 ± 24·3	6·2 ± 15·8	5·1 ± 14·3	1·3 ± 8·4
Dead tree basal area (m^2 ha^{-1})	4·1 ± 8·4	3·2 ± 7·3	2·0 ± 4·9	1·2 ± 3·7
Dead tree density (stems per ha)	120 ± 200	111 ± 209	64 ± 110	38 ± 77
Overstorey tree size diversity (Shannon diversity index)	1·8 ± 1·0	1·6 ± 1·0	1·6 ± 0·9	1·2 ± 0·9
Abiotic				
Elevation (m)	450·3 ± 164·4	267·2 ± 192·8	180·6 ± 121·5	228·8 ± 122·9
Terrain wetness index	−4·8 ± 1·6	−3·7 ± 2·3	−3·7 ± 2·3	−2·7 ± 2·6
Aspect (°)	1·6 ± 31·1	1·8 ± 31·2	1·5 ± 31·1	1·1 ± 26·1
Mean annual temperature of three previous years to the subject year (°C)	5·0 ± 1·5	6·7 ± 1·3	8·5 ± 1·4	8·9 ± 0·7
Mean annual precipitation of three previous years to the subject year (mm)	1309 ± 195	1236 ± 169	1311 ± 169	1117 ± 135
Mean annual vapour pressure of three previous years to the subject year (Pa)	738 ± 65	816 ± 66	903 ± 91	952 ± 49
Mean annual growing degree days (sum of degree days ≥5 °C) of three previous years to the subject year	2651 ± 313	2991 ± 258	3373 ± 356	3410 ± 224
Mean annual frost free days (sum of degree days >0°C) of three previous years to the subject year	2774 ± 316	3121 ± 260	3511 ± 360	3547 ± 224
Deviation from long-term (1980–2014) mean annual temperature to the subject year temperature (°C)	0·09 ± 0·81	0·03 ± 0·75	−0·02 ± 0·69	−0·24 ± 0·71
Deviation from long-term (1980–2014) mean annual precipitation to the subject year precipitation (mm)	52·7 ± 166·7	37·8 ± 177·4	36·4 ± 199·6	8·5 ± 110·5

the climatic data of the three previous years to the measurement year as the presence and abundance of a species in an area represents the cumulative effects of previous years. Daily climate data of 1980–2014 were extracted from ~1 km gridded rasters of Daymet Earth Data (https://daymet.ornl.gov/) using available coordinates of sample plots, whereas topographic wetness index and aspect were extracted from raster layers computed using System for Automated Geoscientific Analyses (SAGA) (Böhner, McCloy & Strobl 2006) (Table 1).

STATISTICAL ANALYSIS

For hypotheses 1 and 2, we modelled the sapling occurrence (presence or absence), absolute (stems per ha) and relative abundance (% of total), respectively, as a function of time, species (four levels) and ecological provinces (four levels). Along with additive effects of time, species and ecological provinces, three-way interactions were also incorporated in the model.

For hypothesis 3, we first developed three candidate models based on (i) biotic variables, (ii) abiotic variables and (iii) biotic and abiotic variables (Table 2). For the biotic model, we considered additive effects of species, live tree stand basal area, dead tree stand basal area, species conspecific live tree basal area (% of total) and two-way interactions among those variables (Table 2). Preliminary analysis indicated that other overstorey variables, including live and dead tree density, and live tree size diversity were highly correlated with live tree stand basal area. Therefore, they were not included in the final model (Table 2). For the abiotic model, we considered four explanatory variables including terrain wetness index, past temperature (mean annual temperature of three previous years), past precipitation (mean total annual precipitation of three previous years) and variability in temperature [deviation from long-term (1980–2014) mean to measurement year]. Other variables were not included considering the both biological assumptions and high correlation with selected variables. In the final abiotic model, we considered the additive effects of species, terrain wetness index, past temperature, past precipitation and variability in temperature, and various two-way interactions. For the third model (biotic and abiotic), we considered all explanatory variables from the prior biotic and abiotic models (Table 2).

Effects of explanatory variables on the distribution of sapling occurrence and relative abundance were assessed by generalized linear mixed-effect models with a binomial distribution, while absolute abundance was fit with Poisson distribution using the function *glmer* in the lme4 package in R (Pinheiro *et al*. 2014; R Development Core Team 2014). Subplots nested within plots and plots nested within measurement years were treated as random factors. Preliminary analysis indicated that an additional error structure to account for plot spatial and temporal autocorrelation did not improve model performance and was not incorporated into the final model. We visually verified the assumptions of normality and variance homogeneity of the residuals, and tested the potential multicollinearity among explanatory variables of a model.

Results

Beech has expanded its occurrence and abundance in the north-eastern US states (Maine, New Hampshire, New York and Vermont), while the presence and abundance of the three other associated deciduous species have either remained stable or decreased. In the BMB forests, the absolute sapling abundance of beech increased almost twofold, while the sapling absolute abundance of sugar maple has decreased (Appendix S7).

SAPLING OCCURRENCE OVER TIME

The presence of beech saplings in forests across the north-eastern US has increased significantly over the past three decades (1983–2014) in three ecological provinces

Table 2. List of candidate models for evaluating the associated factors affecting the occurrence, absolute abundance and the relative abundance of saplings, the results of model performance are based on Akaike Information Criterion (AIC), AIC relative to the most parsimonious model (ΔAIC), area under the curve (AUC) and R^2 of the models

Model type	Model	AIC	ΔAIC	AUC	R^2
Occurrence probability					
Biotic	Y~SPP*BA+SPP*DBA+SPP*CBA	253 802·4	4398·5	0·82	0·15
Abiotic	Y~SPP*MAT+SPP*MAP+SPP*DMAT+SPP*TWI	262 576·5	13 172·6	0·79	0·11
Biotic and Abiotic	Y~SPP*BA+SPP*DBA+SPP*CBA+ SPP*MAT+SPP*MAP+SPP* DMAT+SPP*TWI	249 403·9	–	0·82	0·17
Absolute abundance					
Biotic	Y~SPP*BA+SPP*DBA+SPP*CBA	141 058·0	2017·7	0·76	0·34
Abiotic	Y~SPP*MAT+SPP*MAP+SPP*DMAT+SPP*TWI	149 168·6	10 128·3	0·72	0·27
Biotic and Abiotic	Y~SPP*BA+SPP*DBA+SPP*CBA+ SPP*MAT+SPP*MAP+SPP* DMAT+SPP*TWI	139 040·3	–	0·77	0·35
Relative abundance					
Biotic	Y~SPP*BA+SPP*DBA+SPP*CBA	45 223·0	501·1	0·88	0·22
Abiotic	Y~SPP*MAT+SPP*MAP+SPP*DMAT+SPP*TWI	49 292·2	4570·3	0·51	0·12
Biotic and Abiotic	Y~SPP*BA+SPP*DBA+SPP*CBA+ SPP*MAT+SPP*MAP+SPP* DMAT+SPP*TWI	44 721·9	–	0·87	0·23

SPP, species (beech, sugar maple, red maple and birches); BA, live overstorey basal area; DBA, dead overstorey basal area; CBA, species conspecific basal area (% of total); MAT, mean annual temperature; MAP, mean annual precipitation; DMAT, annual temperature deviation from long-term (1980–2014) mean and TWI, terrain wetness index. SPP*MAT = example of an interaction term.

including Alpine Mixed, North-eastern Mixed and Eastern Broadleaf, but not in the Midwest Broadleaf ecological province. The greatest change occurred in the Alpine Mixed ecological province (Fig. 1a). In contrast, the frequency of occurrence has decreased significantly or remained steady for sugar maple, red maple and birch

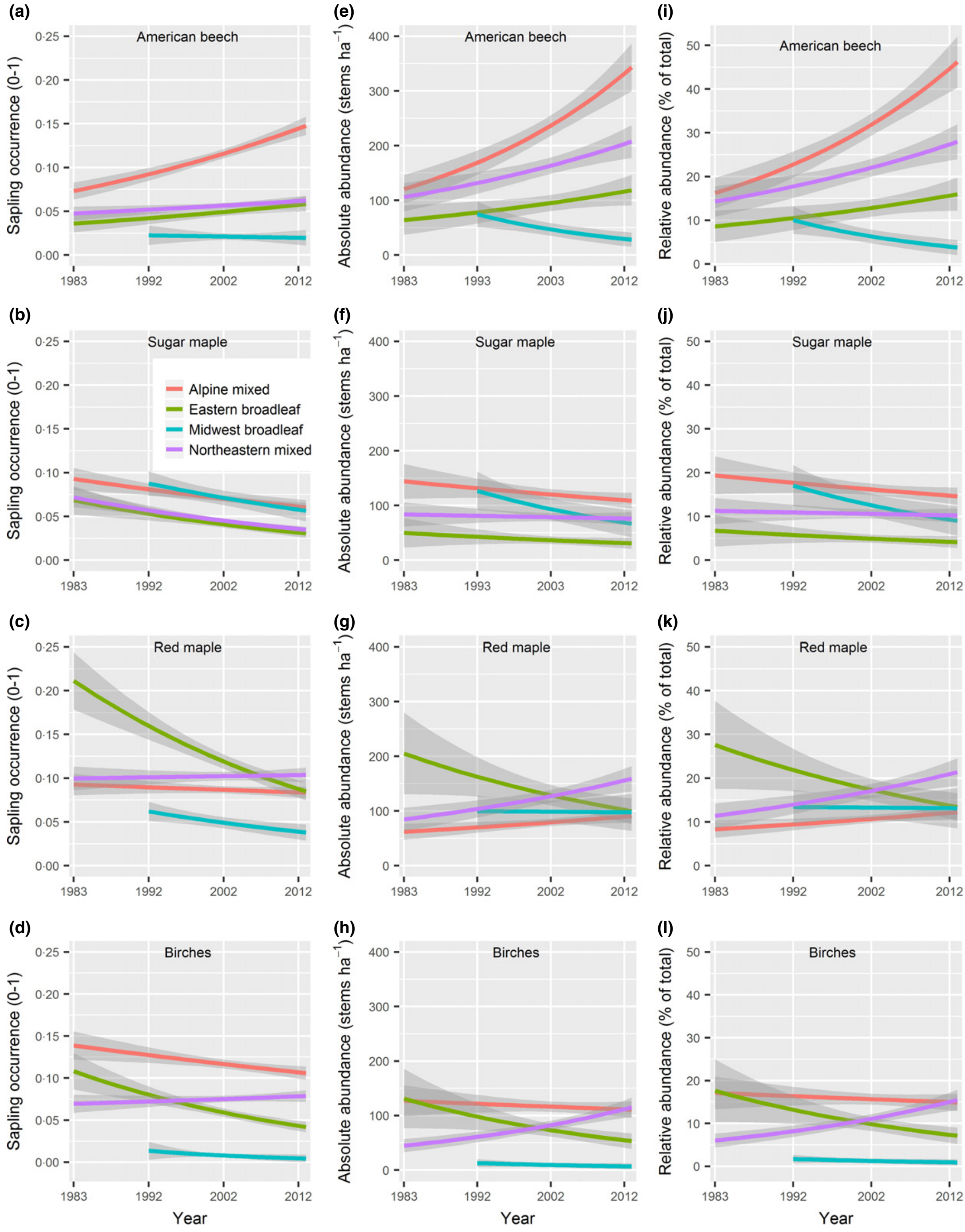

Fig. 1. The occurrence (presence or absence), absolute abundance and relative abundance of saplings (2·5–12·69 cm at dbh) of American beech, sugar maple, red maple and birch species across four ecological provinces of north-eastern USA over the past 32 years. Occurrence was modelled for the entire region, but absolute and relative abundance were modelled only for beech-maple-birch (BMB) forests (≥80% basal area of beech, maple or birch). The solid line is the estimated mean and shaded areas represent the 95% confidence intervals.

across all four ecological provinces. Sapling occurrence for sugar maple, red maple or birch had not increased significantly over the past three decades in any of the ecological provinces (Fig. 1b–d).

SAPLING ABSOLUTE AND RELATIVE ABUNDANCE OVER TIME

In the BMB forests of north-eastern USA, both absolute and relative abundance of beech saplings has increased significantly over the past three decades (1983–2014) in three of four ecological provinces examined including Alpine Mixed, North-eastern Mixed and Eastern Broadleaf, but not in Midwest Broadleaf ecological province. The greatest change occurred in the Alpine Mixed ecological province (Fig. 1e,i). In contrast, sapling abundance in terms of both absolute and relative abundance has decreased significantly for sugar maple in all four ecological provinces as well as for red maple and birch in three ecological provinces, except Midwest Broadleaf. Sapling abundance for sugar maple, red maple and birch did not increase significantly over the last three decades in any ecological province, except for red maple and birch in Midwest Broadleaf ecological province (Fig. 1f–l).

FACTORS AFFECTING THE DISTRIBUTION OF SAPLING OCCURRENCE

The spatial distribution of sapling occurrence over the entire study period was associated with changes in temperature and precipitation, terrain wetness and variability in temperature among abiotic factors, as well as conspecific basal area, stand basal area and dead tree basal area among biotic factors. Of the three candidate models, the model with both biotic and abiotic factors performed best (Table 2) and all parameters were statistically significant. Variability in temperature had a positive effect on the spatial distribution of sapling occurrence for beech, red maple and birch, but a negative effect on sugar maple (Fig. 2a). Past temperature had a unimodal relationship with the spatial distribution of sapling occurrence for all four species (Fig. 2b). Increased precipitation and decreased terrain wetness significantly associated with areas that had higher beech occurrence than three other species (Fig. 2d,f). Beside the conspecific basal area, biotic factors had a weaker association with the spatial distribution of sapling occurrence relative to abiotic factors (Fig. 2e,g). Species conspecific overstorey basal area had a stronger positive association with the spatial distribution of beech sapling occurrence relative to the sapling occurrence of three other species (Fig. 2c).

FACTORS AFFECTING THE DISTRIBUTION OF ABSOLUTE AND RELATIVE ABUNDANCE

As with sapling occurrence, the spatial distribution of sapling abundance over the entire studied period was affected by the past temperature and precipitation, terrain wetness and variability in temperature among abiotic factors, as well as conspecific basal area, stand basal area and dead tree basal area among biotic factors. For all three response variables, the biotic model was superior both in terms of parsimony and fit than the abiotic model. However, the model with both biotic and abiotic variables performed best for both absolute and relative abundance with all parameters being statistically significant (Table 2). Variability in temperature had a strong positive association with the spatial distribution of sapling abundance of beech, but a negative association with the spatial distribution of sugar maple and birch sapling abundance, and no association with red maple (Fig 3a). Past temperature had a unimodal relationship with the spatial distribution of sapling absolute abundance for all four species (Fig. 3b). Increased precipitation and reduced terrain wetness had a strong positive association with the spatial distribution of sapling abundance of beech, but a negative association with the spatial distribution of sapling abundance of the three other species (Fig. 3d,f). Among biotic factors, species conspecific overstorey basal area, stand basal area and dead tree basal area were higher in the areas that had the higher absolute abundance of beech saplings relative to areas with a higher absolute abundance of saplings of the three other species (Fig. 3c,e,g). The response of sapling-relative abundance to all explanatory variables was similar to the responses of sapling absolute abundance for all explanatory variables (Appendix S8).

Discussion

The primary aim of this study were to assess sapling encroachment into new areas as well as the absolute and relative abundance of American beech relative to other commonly associated deciduous species in the north-eastern USA over the past three decades, and relate the spatial distribution of sapling occurrence and abundance with important biotic and abiotic factors. Our results suggest that the sapling occurrence as well as the abundance of American beech have increased over the past three decades (Figs 1 and 4), whereas the sapling occurrence and abundance of three other species (sugar maple, red maple and birch) has decreased. Therefore, a clear shift in sapling tree species composition is currently happening in north-eastern USA, with uncertain consequences for future ecosystem structure and function.

Unlike other dry regions of the world (e.g. southern Europe and southwest USA) (Bernhardt-Römermann *et al.* 2015; Barros *et al.* 2017), where an increased temperature and drought have been identified as important drivers of species distribution, the observed spatial distribution in sapling presence and abundance in the north-eastern USA has likely been driven by a range of factors, including climate (increased precipitation and variability in temperature), disturbance (long absence of wildfire and clear cutting) and species functional characteristics (e.g. shade tolerance).

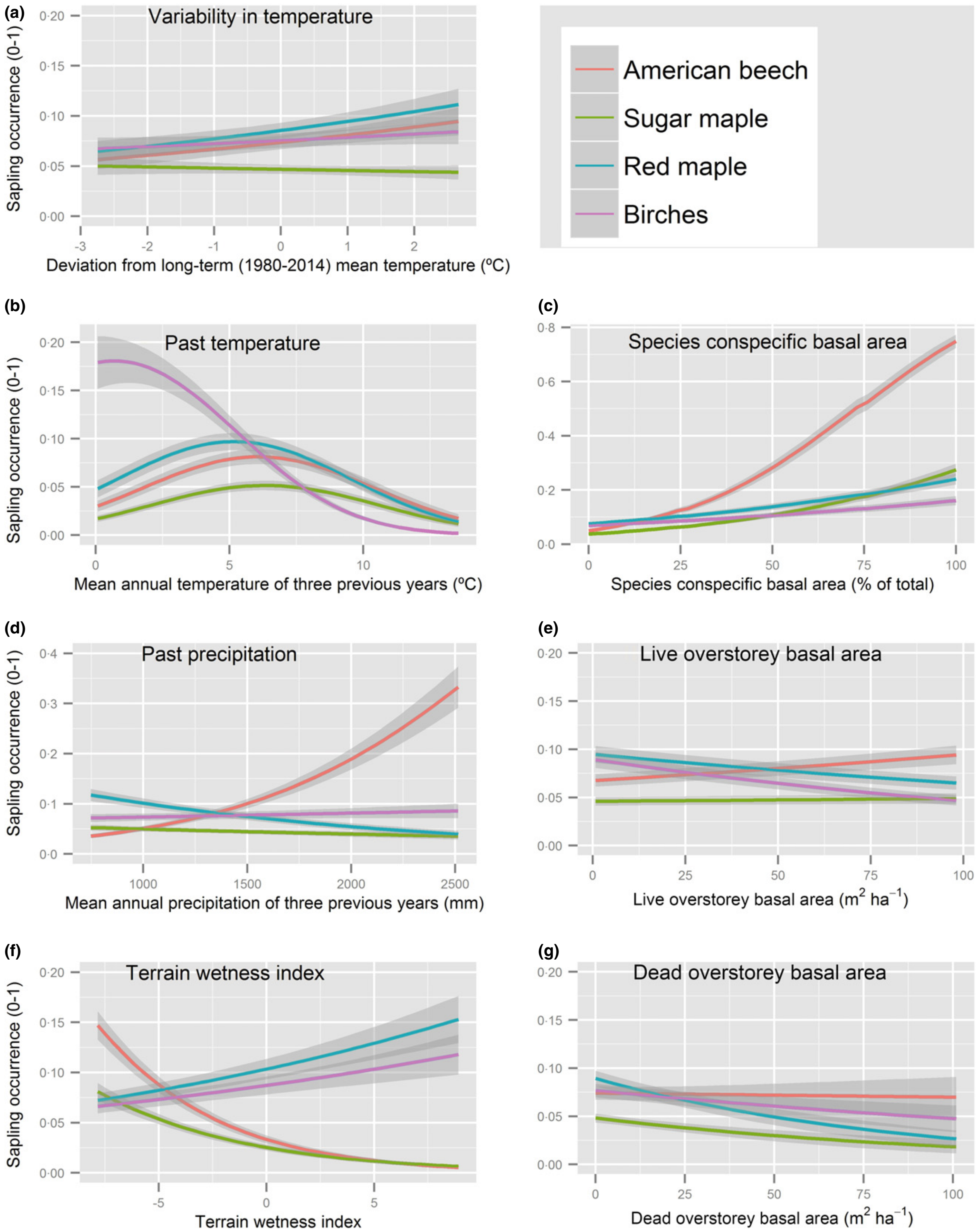

Fig. 2. Abiotic and biotic factors influencing the occurrence (presence or absence) of sapling (2·5–12·69 cm at diameter at breast height) of American beech, sugar maple, red maple and birch species across all forests of north-eastern USA. The solid line is the estimated mean and shaded areas represent the 95% confidence intervals.

Among the driving factors examined, higher precipitation, increased variability in temperature, decreased terrain wetness and greater species conspecific basal area had the strongest effects in discriminating the distribution of beech occurrence as well as abundance against the distribution of the three other hardwood species analysed.

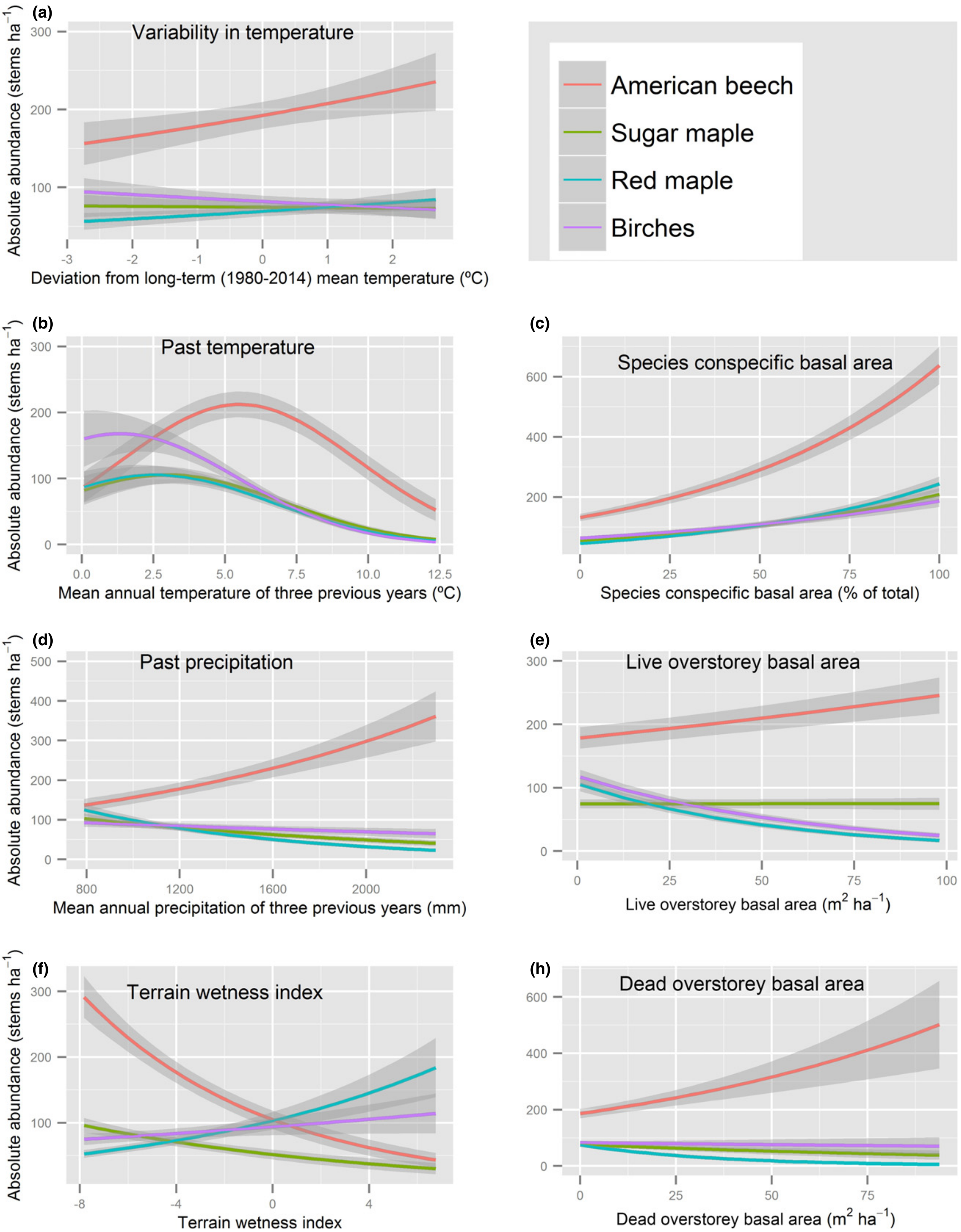

Fig. 3. Abiotic and biotic factors influencing the absolute abundance of sapling (2·5–12·69 cm at dbh) of four major hardwood tree species (American beech, birch, red maple and sugar maple) in the beech-maple-birch (BMB) forests (beech, maples and birches occupied ≥80% of total overstorey basal area) of north-eastern USA. The solid line is the estimated mean and shaded areas represent the 95% confidence intervals.

Despite these important findings, the analysis does have some important limitations including a relatively small plot size (13·5 m^2), large minimum sapling size limit (2·54 cm dbh), limited information on previous disturbance history (Bose *et al.* 2016) and lack of direct measurements of potentially important factors like available radiation (Messier, Parent & Bergeron 1998), water and nutrients (Poorter & Nagel 2000), substrates (Simard, Bergeron & Sirois 1998), herbivore activities (Kuiters & Slim 2002) and below-ground factors (Coomes & Grubb 2000). In addition, the study extracted information on abiotic factors from available spatial datasets, which have varying resolutions of 30–1000 m. This relatively coarse resolution is important when compared to the more precise measurements of the response variables in the permanent plots, which may contribute to the high unexplained variance in the models. Regardless, the coherence of observed and predicted values strongly supports the importance of the outcomes (Fig. 1 and Appendix S7). In addition, the extensive network and consistent measurement protocol allows detection of important landscape trends as observed in this analysis.

TEMPORAL SHIFT

Our results clearly demonstrated that beech has been increasingly encroaching into new forest areas of the north-eastern USA, as well as increasing its abundance locally in the BMB forests (Fig. 4). The result is particularly evident in the Alpine Mixed ecological province, where beech has been expanded its presence and abundance over the past three decades (Fig. 1). The Alpine Mixed ecological province primarily represents the mountainous areas of north-eastern USA, covering the areas of the Adirondack Mountains of New York, the Green Mountains of Vermont, and the White Mountains of New Hampshire (McNab *et al.* 2007). Similarly, Wason & Dovciak (2017) reported beech expansion at higher elevations of north-eastern USA by the reduction in sugar maple and red spruce *Picea rubens* Sarg. This encroachment of beech into new forest areas and increasing abundance in BMB forest areas have not been found to be complementary to the three other associated deciduous species, as their occurrence has generally declined in all four ecological provinces of the north-eastern USA (with few exceptions) (Fig. 1). Although, our study has not analysed saplings of all tree species present, an increasing sapling occurrence of beech in association with a decreasing occurrence of sugar maple, red maple and birch is a clear indicator of a significant vegetation shift in north-eastern USA forests.

ASSOCIATED FACTORS OF THE DISTRIBUTION OF SAPLING PRESENCE AND ABUNDANCE

The presence or absence of a species in an area is determined by species-specific (i) environmental requirements, (ii) dispersal distance to preferred conditions (distance to parents and dispersal agents), (iii) maintenance of a positive growth rate when the presence is true and (iv) survival in competition for resources with neighbours (avoiding competitive exclusion) (Gravel, Guichard & Hochberg 2011; Boulangeat, Gravel & Thuiller 2012). In this study, we analysed well-established and relatively large individuals (2·5–12·69 cm at dbh), and subsequently, the presence of such large sized individual at a location likely indicated their long history of success in meeting all four above-mentioned requirements.

Our results showed that incorporating both abiotic and biotic factors into the same model can improve model performance (parsimony and fit) for sapling occurrence, as well as for absolute and relative abundance. Among abiotic factors, increased precipitation, greater variability in temperature and decreased terrain wetness were associated with the forest areas (FIA plots) that had an increased presence and abundance of beech saplings. Hayhoe *et al.* (2007) demonstrated climate warming and increased precipitation in the north-eastern USA over the past several decades. Our results show that forest areas that received a higher precipitation during the previous 3 years prior to the measurement year had a higher presence and abundance of beech, but a lower presence and abundance of the three other species (except presence of birch) (Figs 2d and 3d), which may indicate that relative to the three other species, beech is more efficient to utilize the growing conditions driven by the increased precipitation. In addition to Hayhoe *et al.* (2007), the Daymet climate estimates used in this analysis also indicated an increased rate of annual precipitation since 2000 in all ecological provinces of north-eastern USA, except Midwest Broadleaf (Appendix S4). This may partially explain why beech occurrence has increased in those three ecological provinces and not in the Midwest Broadleaf (Fig. 1a).

Among biotic factors, species conspecific overstorey basal area had the strongest influence on all response variables examined (Figs 2c and 3c, and Appendix S8C), which indicated that increased propagule availability (greater overstorey basal area) led to higher sapling densities. However, unlike beech, the other species (sugar maple, red maple and birch) have not responded linearly to the increased conspecific basal area. Therefore, overstorey presence or the presence of parent trees has a limited influence on sugar maple, red maple and birch saplings. This is due, at least partially, to: (i) sugar maple, red maple and birch are more shade-intolerant species relative to beech, hence regeneration growth and survival are more related to understorey light availability than the availability of parent trees (Kobe *et al.* 1995), (ii) browsers (white-tailed deer *Odocoileus virginianus* Zimmerman and moose *Alces alces* L.) have preference for sugar maple, red maple and birch over beech, and (iii) species shade tolerance behaviour is directly linked to the species assimilation potential during photosynthesis (i.e. light use efficiency). Consequently, overstorey sugar maple, red maple and birch may generate

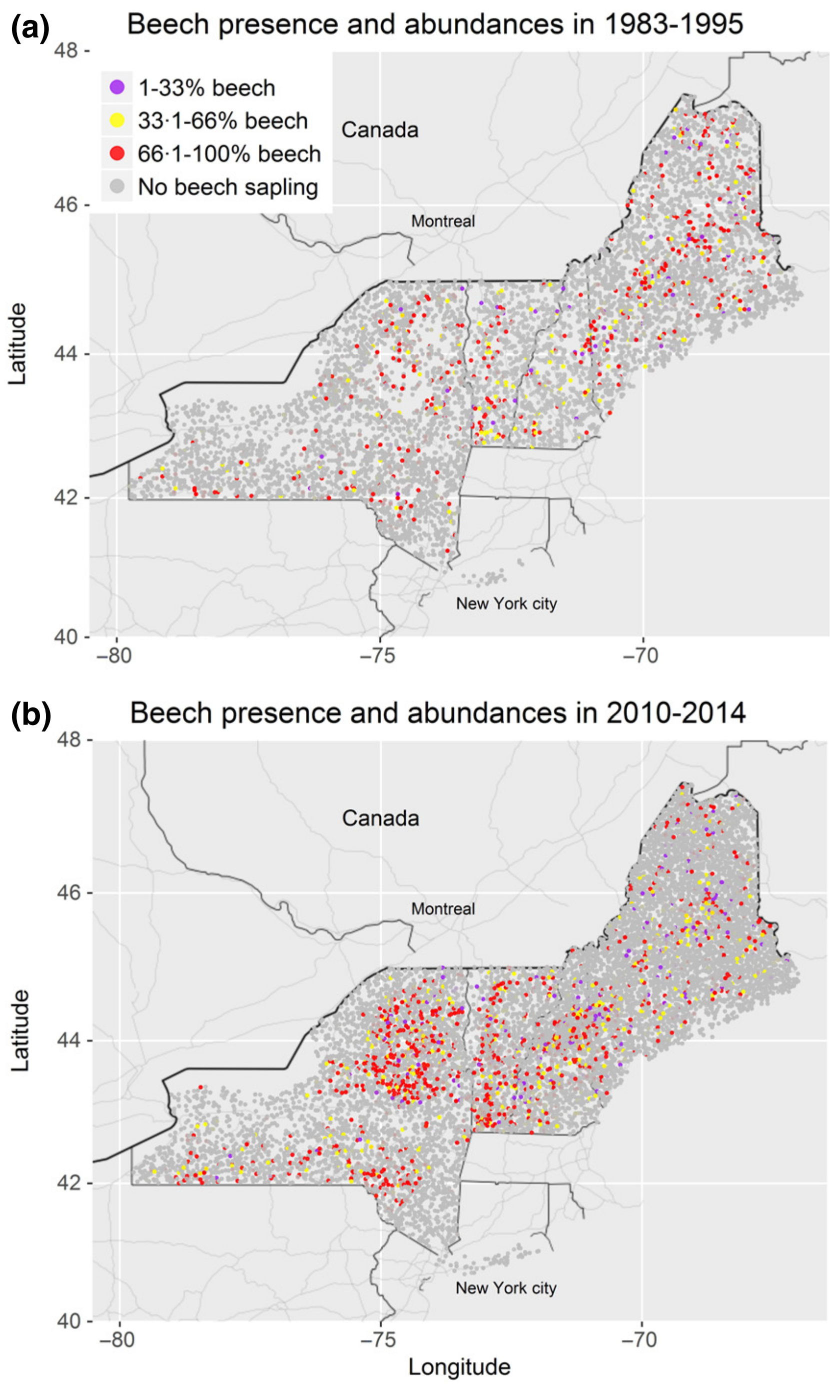

Fig. 4. Spatial changes over the past three decades in sapling (2·5–11·7 cm at DBH) presence and abundance of American beech across the forested landscape of north-eastern USA (Maine, New Hampshire, New York and Vermont). (a) Beech presence and abundances in 1983–1995, and (b) Beech presence and abundances in 2010–2014.

greater competition to their respective understorey saplings relative to beech (Thomas *et al*. 1999).

The biotic interactions among multiple species drive species coexistence or competitive exclusion (Boulangeat, Gravel & Thuiller 2012). In our study, the interaction was apparently negative (positive competitive exclusion), where an increasing dead tree basal area positively influenced the abundance of beech, but was negatively related to the abundance of other deciduous species (Fig. 2g and Appendix S8G). When a beech sapling was present in the understorey, overstorey tree mortality (creation of canopy gaps) benefited the beech saplings more than the three other deciduous species (sugar maple, red maple, and birch).

Negative interaction (competitive exclusion) generally occurs when multiple species compete for resources (Chase & Leibold 2003). In the forests of north-eastern USA, several factors have been reported as beneficial to beech with respect to competition with sugar maple, red maple and birch, namely: (i) beech is more shade tolerant than the three other studied species (Canham 1989), therefore, small-scale overstorey openings that occur by natural mortality due to beech-bark disease, self-thinning and senescence, and by selection harvest are more beneficial to beech saplings than other species, (ii) unlike the three other species, beech is able to regenerate vegetatively from root suckers, which is often promoted by the beech-bark disease and root injuries during harvesting (Houston 2001; Nyland *et al.* 2006). Beech seedlings originated from suckering have shown a higher rate of growth and survivability than beech seedlings originated from seed (Beaudet & Messier 2008), (iii) beech is a less palatable species for browsers (Tripler *et al.* 2005), and (iv) beech forms an interconnected root system, which effectively generates higher survivability of beech regeneration than other hardwood species (Nyland *et al.* 2006).

CLIMATE CHANGE, BEECH EXPANSION, AND MANAGEMENT IMPLICATIONS

The north-eastern region of USA has experienced increases in temperature of 0·25 °C per decade since 1970, with associated increases in winter temperature and annual precipitation (DeGaetano & Allen 2002; Hayhoe *et al.* 2007), and these changes are projected to continue to increase over the next 100 years (Hayhoe *et al.* 2007). The increased temperature will also result in higher variability in annual temperature from the long-term average, concomitantly drive changes in the growing conditions for tree species. Our results, as well as those of other studies (e.g, Reid & Strain 1994; Fang & Lechowicz 2006), show that climate-change driven outcomes (e.g. greater precipitation, increased growing season temperature, higher atmospheric CO_2 concentration and increased variability in temperature) are associated with increasing beech regeneration in the north-eastern USA relative to other hardwood species.

Our results suggest an ongoing compositional shift in the forests of north-eastern USA over the past three decades, with a general decline of sugar maple, red maple and birch saplings with an increasing American beech sapling density. Since beech is increasingly dominating the sapling layer of forests of north-eastern USA, the change from BMB forests to more beech-dominated forests may occur across even greater areas if large-scale harvesting/disturbances (i.e. large-scale canopy opening) do not occur. The current expansion of beech may also increase the risk of beech-bark disease expansion and reduce the overall productivity of north-eastern forests. Therefore, forest managers will need to incorporate these changes into their long-term management planning. Where beech has become dense in the understorey of northern hardwood forests, no canopy light manipulation strategy by harvesting will succeed in establishing regeneration of desirable species (Nyland *et al.* 2006). Under such situations, regeneration control treatments (e.g. herbicides, girdling, cutting) must remove the undesirable beech prior to harvesting the overstorey. The choice of regeneration control treatments can be based on the magnitude of beech presence (stems per ha), the size of the area and the associated costs. For example, stem girdling can be used in stands have relatively lower abundance of beech (e.g. ~1000 stems per ha), while herbicide spray can be used in stands with greater abundance of beech (>1000 stems per ha) (Nyland *et al.* 2006). In addition, protecting a regeneration area from browsing and conducting logging in winter (when the soil is covered by snow) to minimize root injury induced beech suckering will benefit the regeneration of desirable species (i.e. sugar maple and birch) (Tripler *et al.* 2005; Nyland *et al.* 2006; Wagner *et al.* 2010).

Authors' contributions

A.K.B. and A.W. conceived the ideas and designed methodology; A.K.B. analysed the data, and led the writing of the manuscript with the important contributions of A.W. and R.G.W.

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Data accessibility

A data file named 'Hardwood sapling composition of north-eastern forests' and an instruction file are available from Dryad Digital Repository https://doi.org/10.5061/dryad.qj8gh (Bose, Weiskittel & Wagner 2017).

References

Allen, C.D., Macalady, A.K., Chenchouni, H. *et al.* (2010) A global overview of drought and heat-induced tree mortality reveals emerging climate change risks for forests. *Forest Ecology and Management*, **259**, 660–684.

Barros, C., Guéguen, M., Douzet, R., Carboni, M., Boulangeat, I., Zimmermann, N.E., Münkemüller, T. & Thuiller, W. (2017) Extreme climate events counteract the effects of climate and land-use changes in Alpine tree lines. *Journal of Applied Ecology*, **54**, 39–50.

Beaudet, M. & Messier, C. (2008) Beech regeneration of seed and root sucker origin: a comparison of morphology, growth, survival, and response to defoliation. *Forest Ecology and Management*, **255**, 3659–3666.

Bellard, C., Bertelsmeier, C., Leadley, P., Thuiller, W. & Courchamp, F. (2012) Impacts of climate change on the future of biodiversity. *Ecology Letters*, **15**, 365–377.

Bernhardt-Römermann, M., Baeten, L., Craven, D. *et al.* (2015) Drivers of temporal changes in temperate forest plant diversity vary across spatial scales. *Global Change Biology*, **21**, 3726–3737.

Bertrand, R., Lenoir, J., Piedallu, C., Riofrío-Dillon, G., de Ruffray, P., Vidal, C., Pierrat, J.-C. & Gégout, J.-C. (2011) Changes in plant community composition lag behind climate warming in lowland forests. *Nature*, **479**, 517–520.

Böhner, J., McCloy, K.R. & Strobl, J. (2006) SAGA – analysis and modelling applications. *Göttinger Geographische Abhandlungen*, **115**, 130.

Boisvenue, C. & Running, S.W. (2006) Impacts of climate change on natural forest productivity – evidence since the middle of the 20th century. *Global Change Biology*, **12**, 862–882.

Bose, A.K., Weiskittel, A., Wagner, R.G. & Kuehne, C. (2016) Assessing the factors influencing natural regeneration patterns in the diverse, multi-cohort, and managed forests of Maine, USA. *Journal of Vegetation Science*, **27**, 1140–1150.

Bose, A.K., Weiskittel, A. & Wagner, R.G. (2017) Data from: A three decade assessment of climate-associated changes in forest composition across the north-eastern USA. *Dryad Digital Repository*, https://doi.org/10.5061/dryad.qj8gh

Boulangeat, I., Gravel, D. & Thuiller, W. (2012) Accounting for dispersal and biotic interactions to disentangle the drivers of species distributions and their abundances. *Ecology Letters*, **15**, 584–593.

Canham, C.D. (1989) Different responses to gaps among shade-tolerant tree species. *Ecology*, **70**, 548–550.

Cavin, L. & Jump, A. (2017) Highest drought sensitivity and lowest resistance to growth suppression is found in the range core of the tree *Fagus sylvatica* L. not the equatorial range edge. *Global Change Biology*, **23**, 362–379.

Chase, J.M. & Leibold, M.A. (2003) *Ecological Niches*. Chicago University Press, Chicago, IL, USA.

Coomes, D.A. & Grubb, P.J. (2000) Impacts of root competition in forests and woodlands: a theoretical framework and review of experiments. *Ecological Monographs*, **70**, 171–207.

DeGaetano, A.T. & Allen, R.J. (2002) Trends in twentieth-century temperature extremes across the United States. *Journal of Climate*, **15**, 3188–3205.

Devi, N., Hagedorn, F., Moiseev, P., Bugmann, H., Shiyatov, S., Mazepa, V. & Rigling, A. (2008) Expanding forests and changing growth forms of Siberian larch at the Polar Urals treeline during the 20th century. *Global Change Biology*, **14**, 1581–1591.

Duchesne, L. & Ouimet, R. (2009) Present-day expansion of American beech in northeastern hardwood forests: does soil base status matter? *Canadian Journal of Forest Research*, **39**, 2273–2282.

Fang, J. & Lechowicz, M.J. (2006) Climatic limits for the present distribution of beech (*Fagus* L.) species in the world. *Journal of Biogeography*, **33**, 1804–1819.

Fournier, B., Mouquet, N., Leibold, M.A. & Gravel, D. (2016) An integrative framework of coexistence mechanisms in competitive metacommunities. *Ecography*, in press.

Gravel, D., Guichard, F. & Hochberg, M.E. (2011) Species coexistence in a variable world. *Ecology Letters*, **14**, 828–839.

Grubb, P.J. (1977) The maintenance of species-richness in plant communities: the importance of the regeneration niche. *Biological Reviews of the Cambridge Philosophical Society*, **52**, 107–145.

Hane, E.N. (2003) Indirect effects of beech bark disease on sugar maple seedling survival. *Canadian Journal of Forest Research*, **33**, 807–813.

Hansen, A.J., Neilson, R.P., Dale, V.H., Flather, C.H., Iverson, L.R., Currie, D.J., Shafer, S., Cook, R. & Bartlein, P.J. (2001) Global change in forests: responses of species, communities, and biomes: interactions between climate change and land use are projected to cause large shifts in biodiversity. *BioScience*, **51**, 765–779.

Harsch, M.A., Hulme, P.E., McGlone, M.S. & Duncan, R.P. (2009) Are treelines advancing? A global meta analysis of treeline response to climate warming. *Ecology Letters*, **12**, 1040–1049.

Hayhoe, K., Wake, C.P., Huntington, T.G. *et al.* (2007) Past and future changes in climate and hydrological indicators in the US Northeast. *Climate Dynamics*, **28**, 381–407.

Houston, D.R. (2001) Effect of harvesting regime on beech root sprouts and seedlings in a North-Central Maine forest long affected by beech bark disease. Research Paper NE-717. USDA Forest Service, Newtoin Square, PA, USA.

Hubbell, S.P. (2001) *The Unified Neutral Theory of Biodiversity and Biogeography*. Princeton University Press, Princeton, NJ, USA.

Kobe, R.K., Pacala, S.W., Silander, J.A. Jr & Canham, C.D. (1995) Juvenile tree survivorship as a component of shade tolerance. *Ecological Applications*, **5**, 517–532.

Kuiters, A. & Slim, P. (2002) Regeneration of mixed deciduous forest in a Dutch forest-heathland, following a reduction of ungulate densities. *Biological Conservation*, **105**, 65–74.

Lenoir, J., Gégout, J.-C., Dupouey, J.-L., Bert, D. & Svenning, J.C. (2010) Forest plant community changes during 1989-2007 in response to climate warming in the Jura Mountains (France and Switzerland). *Journal of Vegetation Science*, **21**, 949–964.

Lenoir, J. & Svenning, J.-C. (2015) Climate-related range shifts – a global multidimensional synthesis and new research directions. *Ecography*, **38**, 15–28.

Manning, A.D., Gibbons, P. & Lindenmayer, D.B. (2009) Scattered trees: a complementary strategy for facilitating adaptive responses to climate change in modified landscapes? *Journal of Applied Ecology*, **46**, 915–919.

McNab, W.H., Cleland, D.T., Freeouf, J.A., KeysJr, J.E., Nowacki, G.J. & Carpenter, C.A. (2007) Description of ecological subregions: sections of the conterminous United States. Tech. Rep. WO-76B [CD-ROM]. United States Department of Agriculture, Forest Service, Washington, DC, USA.

Messier, C., Parent, S. & Bergeron, Y. (1998) Effects of overstory and understory vegetation on the understory light environment in mixed boreal forests. *Journal of Vegetation Science*, **9**, 511–520.

Nyland, R.D., Bashant, A.L., Bohn, K.K. & Verostek, J.M. (2006) Interference to hardwood regeneration in northeastern North America: controlling effects of American beech, striped maple, and hobblebush. *Northern Journal of Applied Forestry*, **23**, 122–132.

Parmesan, C. (2006) Ecological and evolutionary responses to recent climate change. *Annual Review of Ecology, Evolution, and Systematics*, **37**, 637–669.

Patterson, S.L., Zak, D.R., Burton, A.J., Talhelm, A.F. & Pregitzer, K.S. (2012) Simulated N deposition negatively impacts sugar maple regeneration in a northern hardwood ecosystem. *Journal of Applied Ecology*, **49**, 155–163.

Perry, D.A., Oren, R. & Hart, S.C. (2008) *Forest Ecosystems*. The Johns Hopkins University Press, Baltimore, MD, USA.

Pinheiro, J., Bates, D., DebRoy, S. & Sarkar, D. (2014) nlme: Linear and nonlinear mixed effects models. R package version 3.1-117. Available at: https://cran.r-project.org/web/packages/nlme/index.html (accessed 10 April 2016).

Poorter, H. & Nagel, O. (2000) The role of biomass allocation in the growth response of plants to different levels of light, CO_2, nutrients and water: a quantitative review. *Functional Plant Biology*, **27**, 1191.

R Development Core Team (2014) *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. Available at: www.r-project.org (accessed 10 April 2016).

Reid, C.D. & Strain, B.R. (1994) Effects of CO_2 enrichment on whole-plant carbon budget of seedlings of *Fagus grandifolia* and *Acer saccharum* in low irradiance. *Oecologia*, **98**, 31–39.

Richardson, A.D., Bailey, A.S., Denny, E.G., Martin, C.W. & O'Keefe, J. (2006) Phenology of a northern hardwood forest canopy. *Global Change Biology*, **12**, 1174–1188.

Rigling, A., Bigler, C., Eilmann, B. *et al.* (2013) Driving factors of a vegetation shift from Scots pine to pubescent oak in dry Alpine forests. *Global Change Biology*, **19**, 229–240.

Simard, M.-J., Bergeron, Y. & Sirois, L. (1998) Conifer seedling recruitment in a southeastern Canadian boreal forest: the importance of substrate. *Journal of Vegetation Science*, **9**, 575–582.

Thomas, S.C., Halpern, C.B., Falk, D.A., Liguori, D.A. & Austin, K.A. (1999) Plant diversity in managed forests: understory responses to thinning and fertilization. *Ecological Applications*, **9**, 864–879.

Thuiller, W., Araujo, M.B. & Lavorel, S. (2004) Do we need land-cover data to model species distributions in Europe? *Journal of Biogeography*, **31**, 353–361.

Thuiller, W., Münkemüller, T., Schiffers, K.H. *et al.* (2014) Does probability of occurrence relate to population dynamics? *Ecography*, **37**, 1155–1166.

Tripler, C.E., Canham, C.D., Inouye, R.S. & Schnurr, J.L. (2005) Competitive hierarchies of temperate tree species: interactions between resource availability and white-tailed deer. *Ecoscience*, **12**, 494–505.

Wagner, S., Collet, C., Madsen, P., Nakashizuka, T., Nyland, R.D. & Sagheb-Talebi, K. (2010) Beech regeneration research: from ecological to silvicultural aspects. *Forest Ecology and Management*, **259**, 2172–2182.

Wason, J.W. & Dovciak, M. (2017) Tree demography suggests multiple directions and drivers for species range shifts in mountains of Northeastern United States. *Global Change Biology*, **23**, 3335–3347.

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Supporting Information

Details of electronic Supporting Information are provided below.

Appendix S1. Description of study area.

Appendix S2. Description of four ecological provinces of north-eastern USA.

Appendix S3. The map of four ecological provinces across four states (Maine, New Hampshire, New York and Vermont) of north-eastern USA.

Appendix S4. The mean daily temperature (°C) and total mean annual precipitation (mm) over the last 34 years (1980–2014) of four ecological provinces of north-eastern USA. Note. solid lines represent the mean, whereas dashed lines represent ± 1 SD from the mean.

Appendix S5. Description of data.

Appendix S6. Summary of sample plots with the history of measurements and the total number of plots and beech-maple-birch (BMB) plots (beech, maples, and birches occupied ≥80% of total overstorey basal area) occupied by four ecological provinces examined.

Appendix S7. Observed sapling occurrence in the entire north-eastern USA by ecological provinces, and observed sapling absolute and relative abundance in beech-maple-birch (BMB) forests (beech, maples and birches occupied ≥80% of total overstorey basal area) of north-eastern USA of four deciduous species over the two-time periods (1983/1993 and 2010–2014). For the absolute and relative sapling abundance mean ± 1 SD were presented.

Appendix S8. Abiotic and biotic factors drive relative abundance of sapling (2·5–12·69 cm at dbh) of four major deciduous tree species in the beech-maple-birch (BMB) forests (beech, maples and birches occupied ≥80% of total overstorey basal area) of north-eastern forests of USA. The solid line is the estimated mean and shaded areas represent the 95% confidence intervals.